

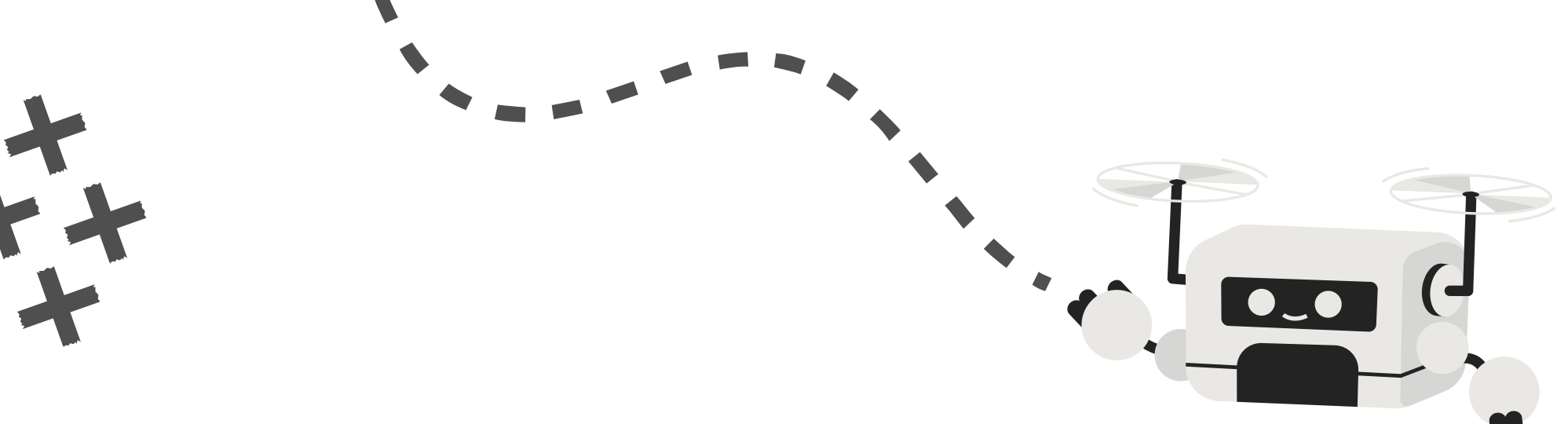
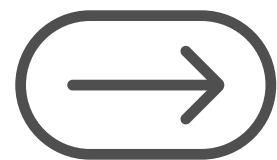
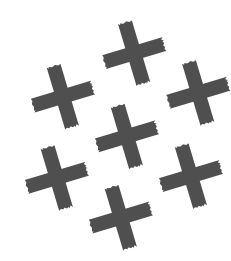
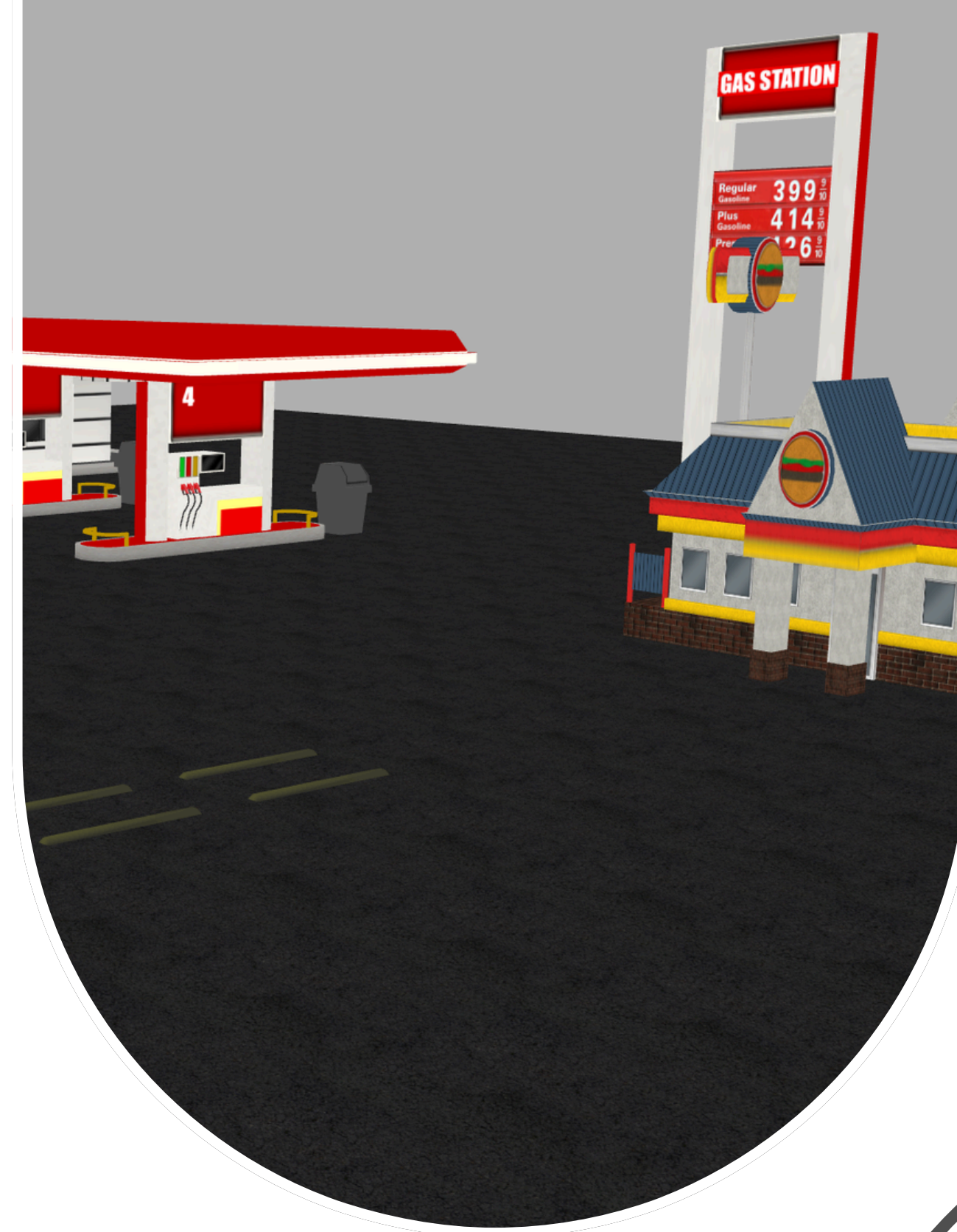


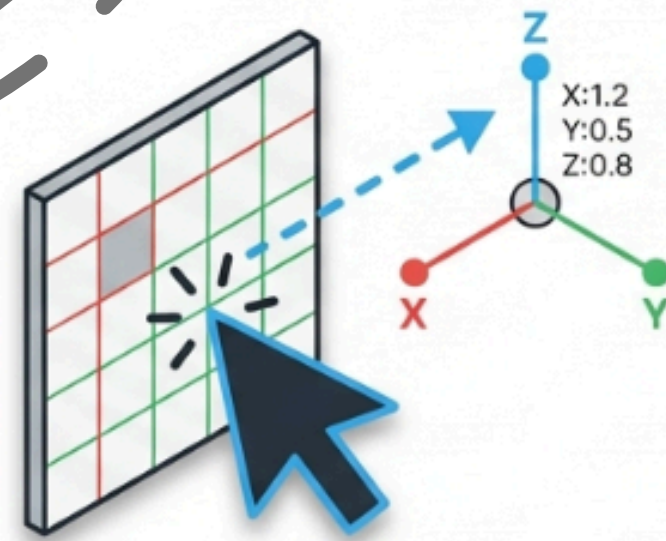
Image-Based Motion Control for a Drone



Presented By:
**Lucia Gumbau
Doñate**

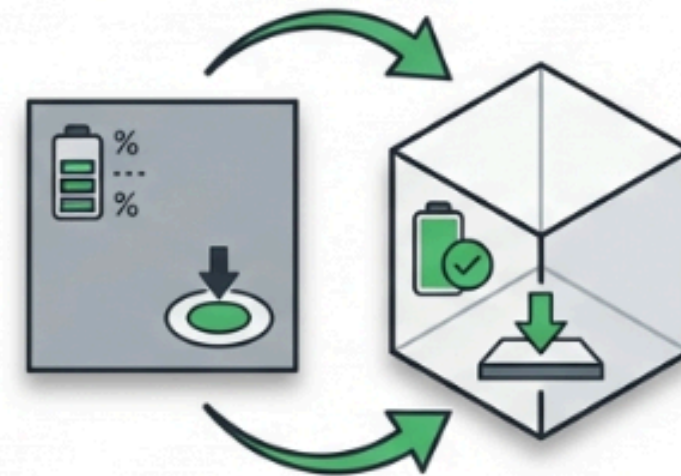


Objectives



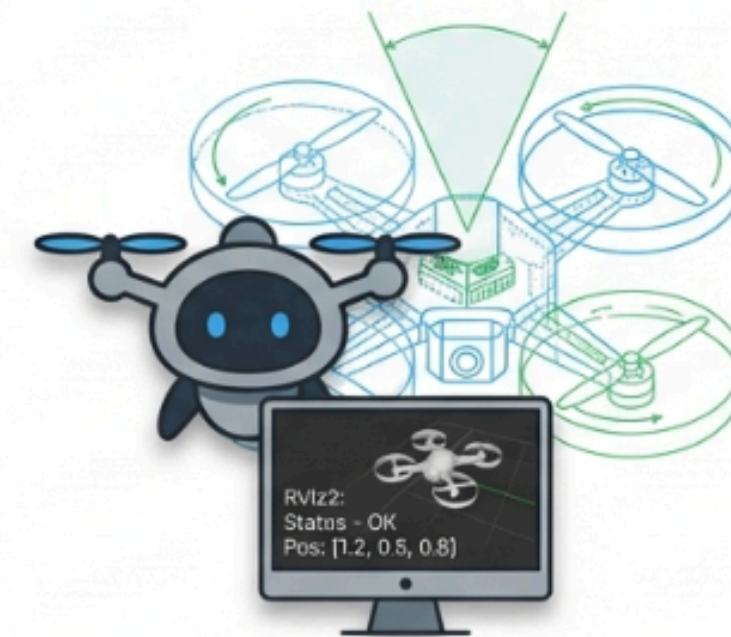
Point-and-Click Navigation

Translating 2D image pixels to 3D spatial coordinates using RGB-D data.



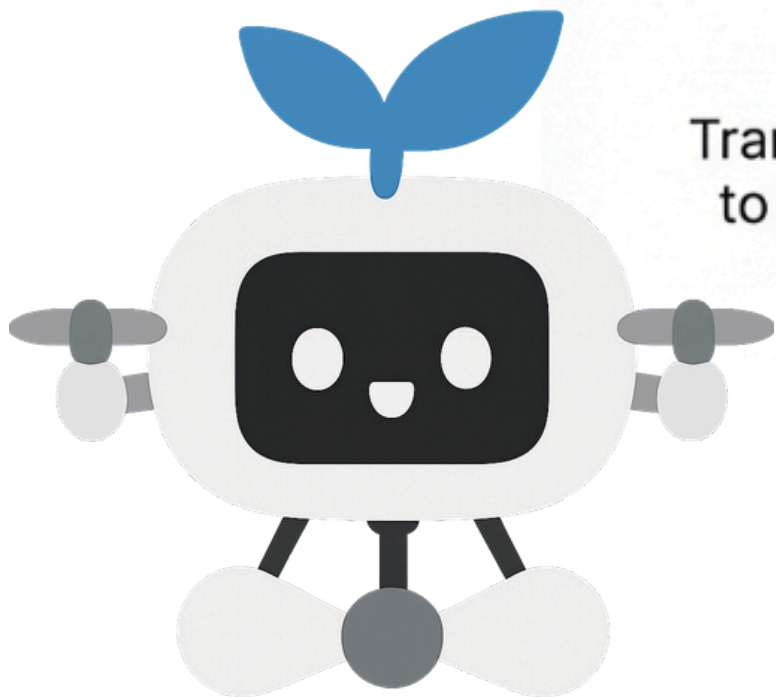
Safety Logic

Autonomous battery management and safe landing zone detection.

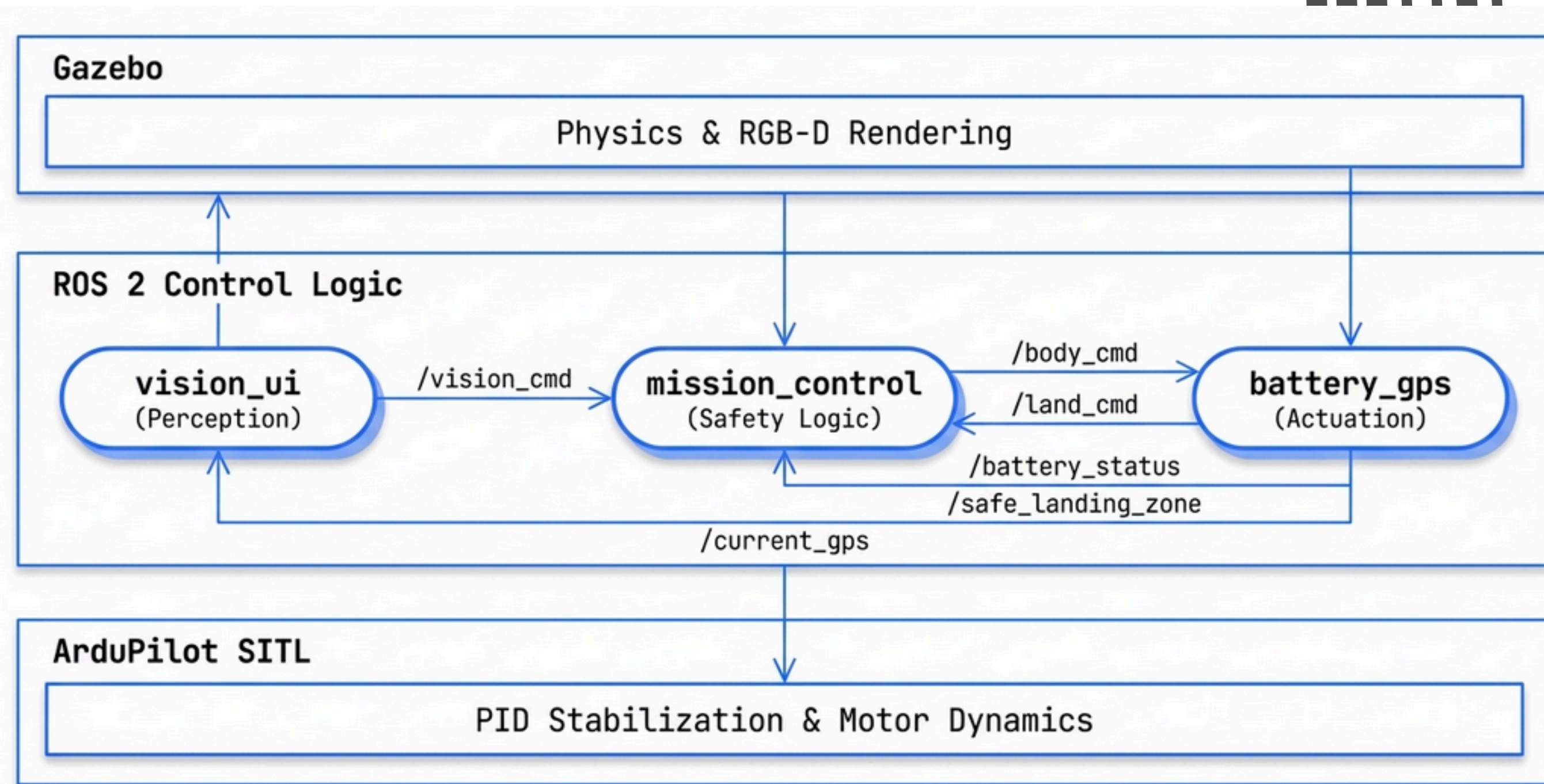


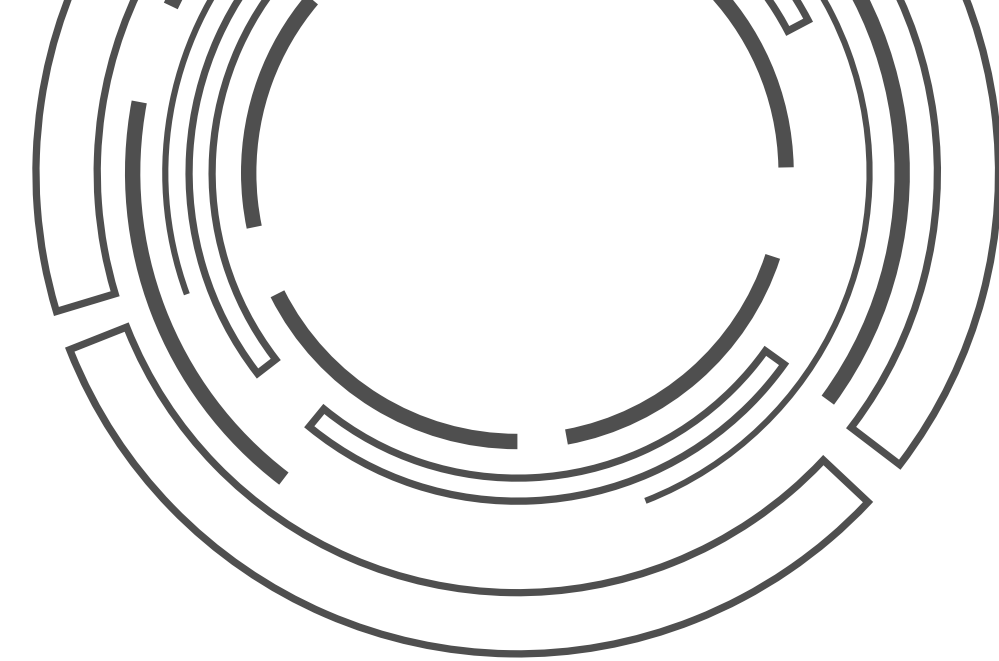
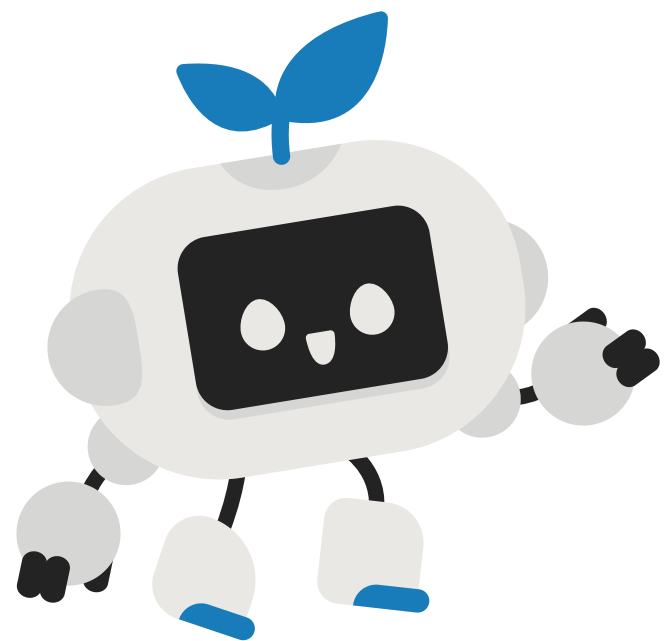
Digital Twin

Real-time 3D monitoring via RViz2.

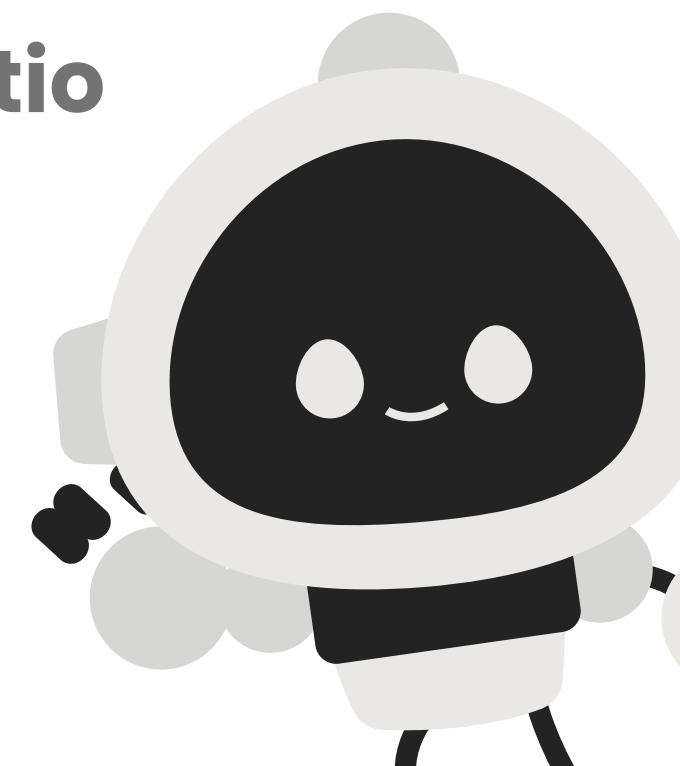
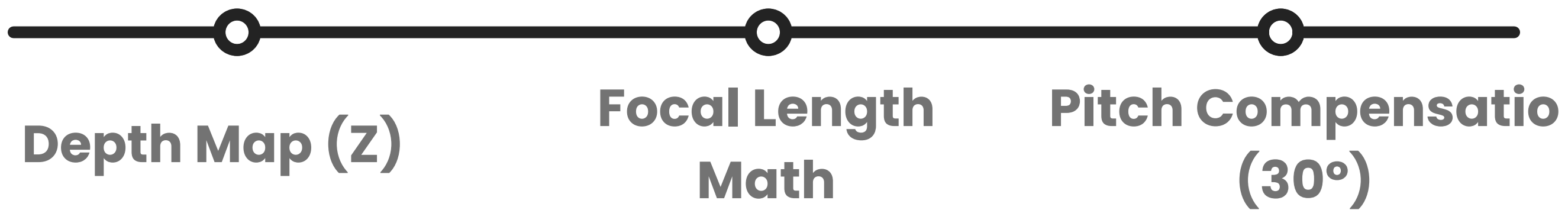


System architecture

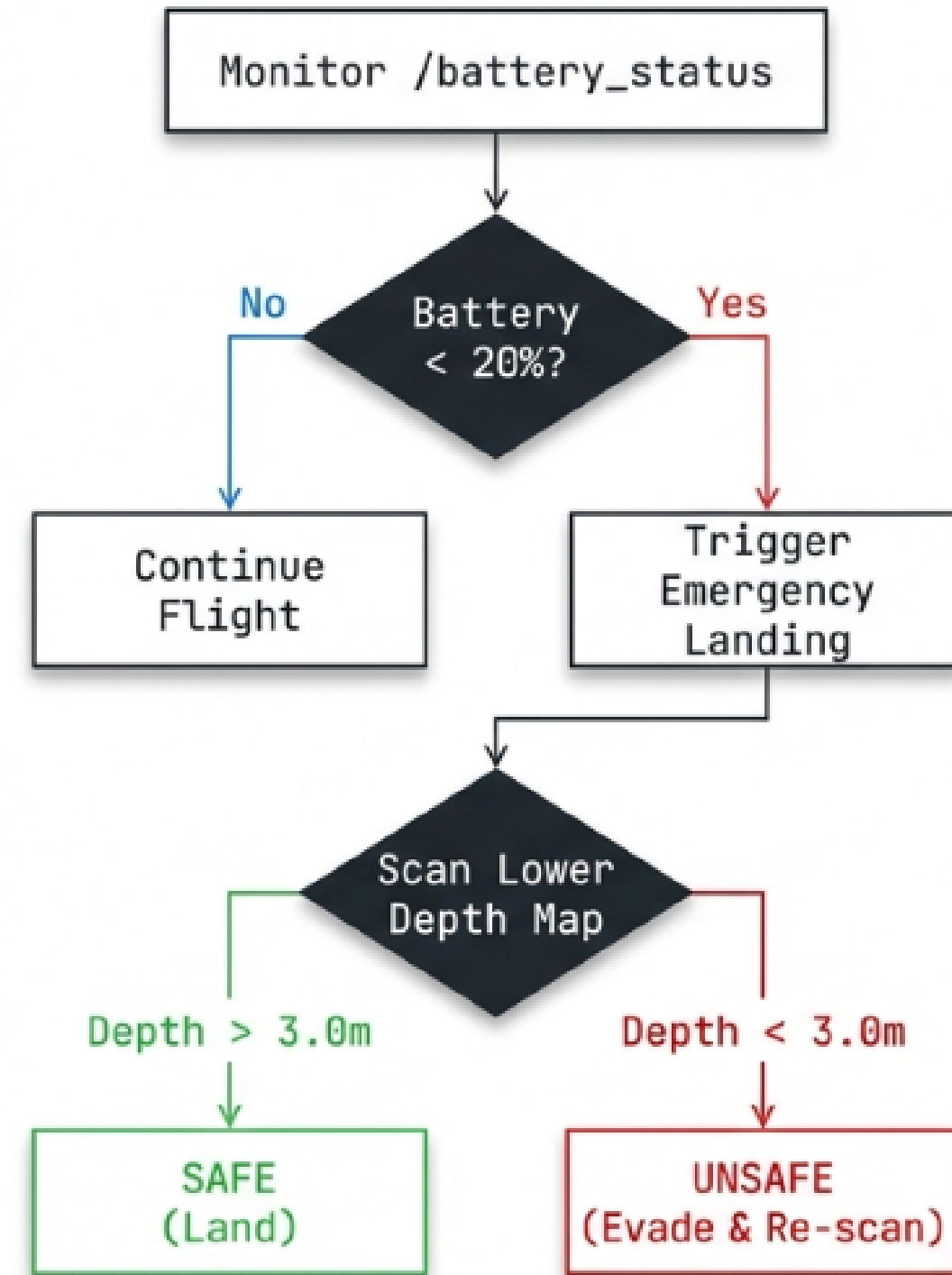
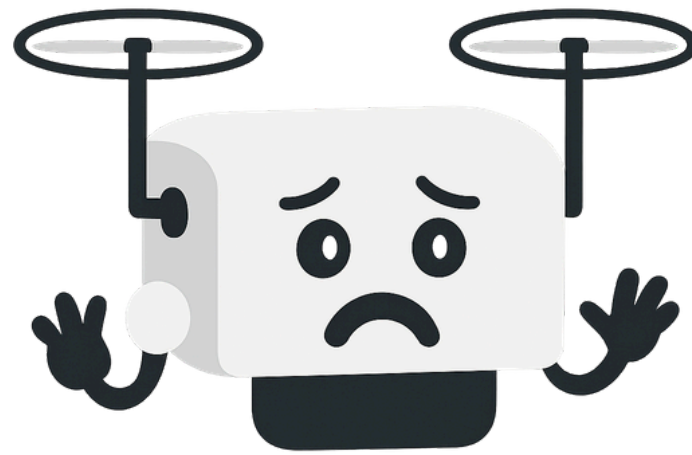
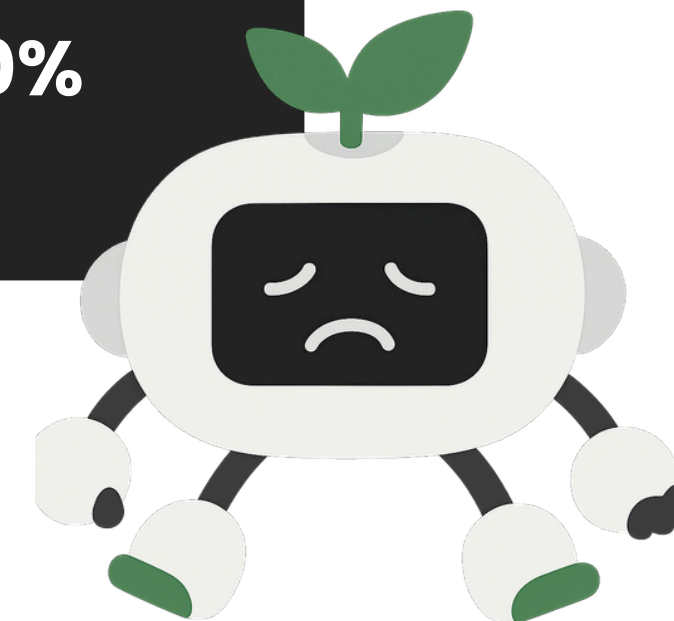
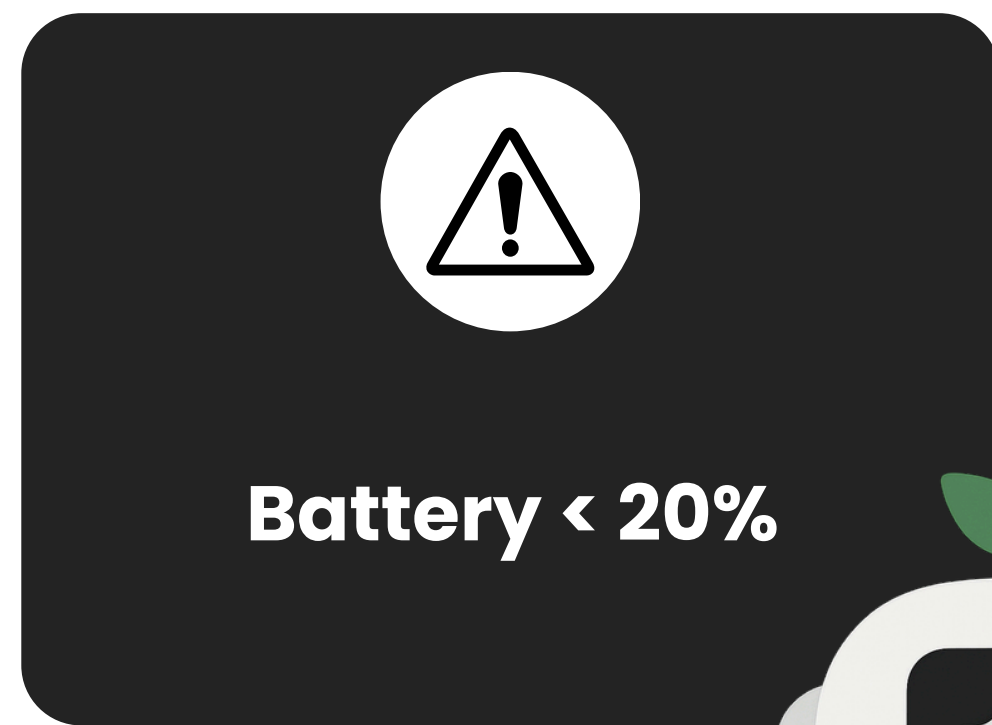




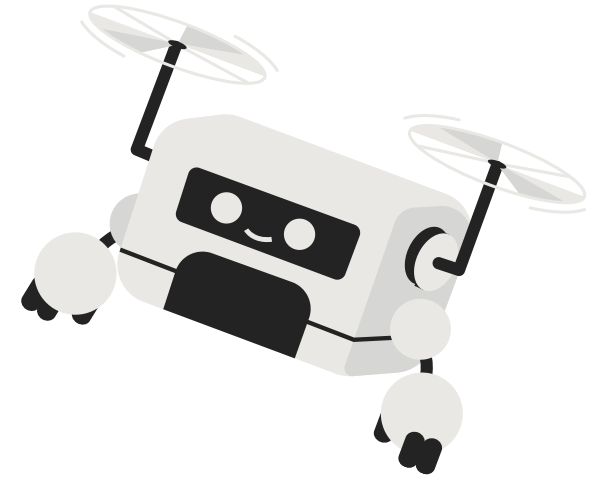
From 2D pixel to 3D Movement



Logic for safety



Flight Execution and MAVLink



0b110**1111**1000

Locks Yaw Rotation
(Prevents Spinning)

Node: battery_gps.py

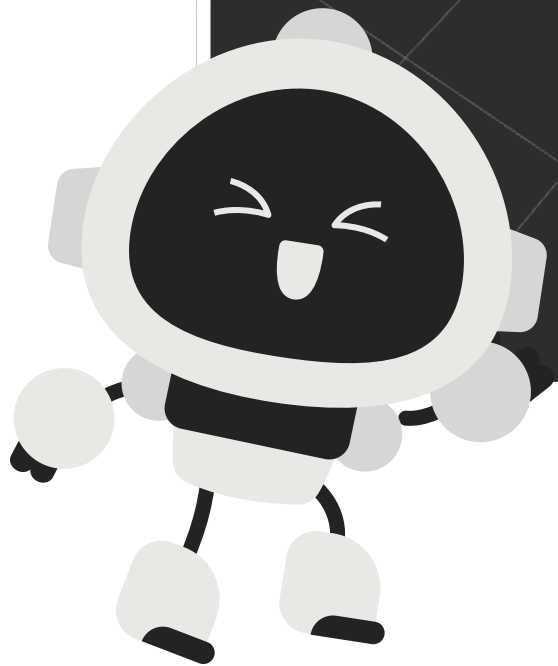
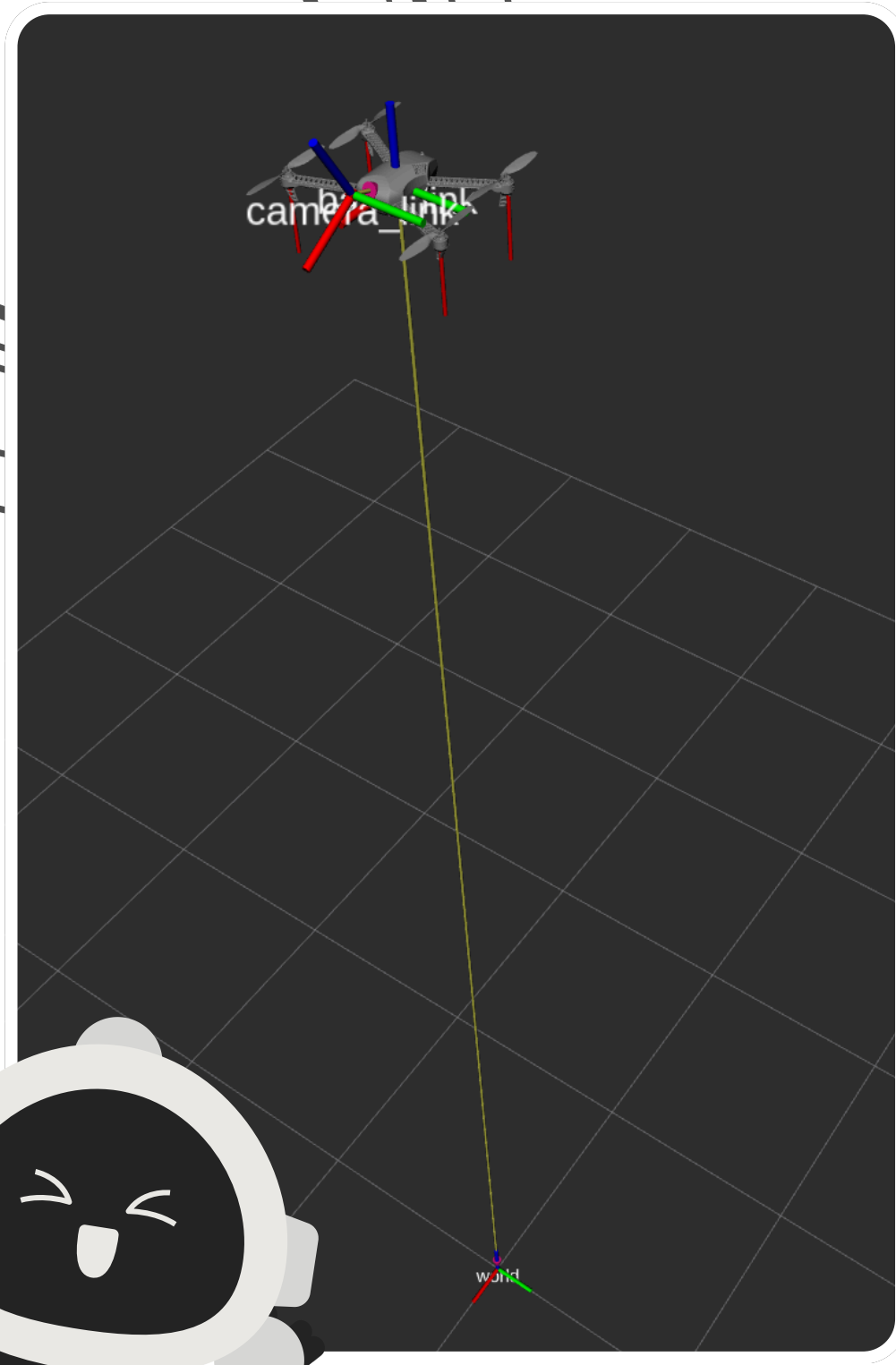
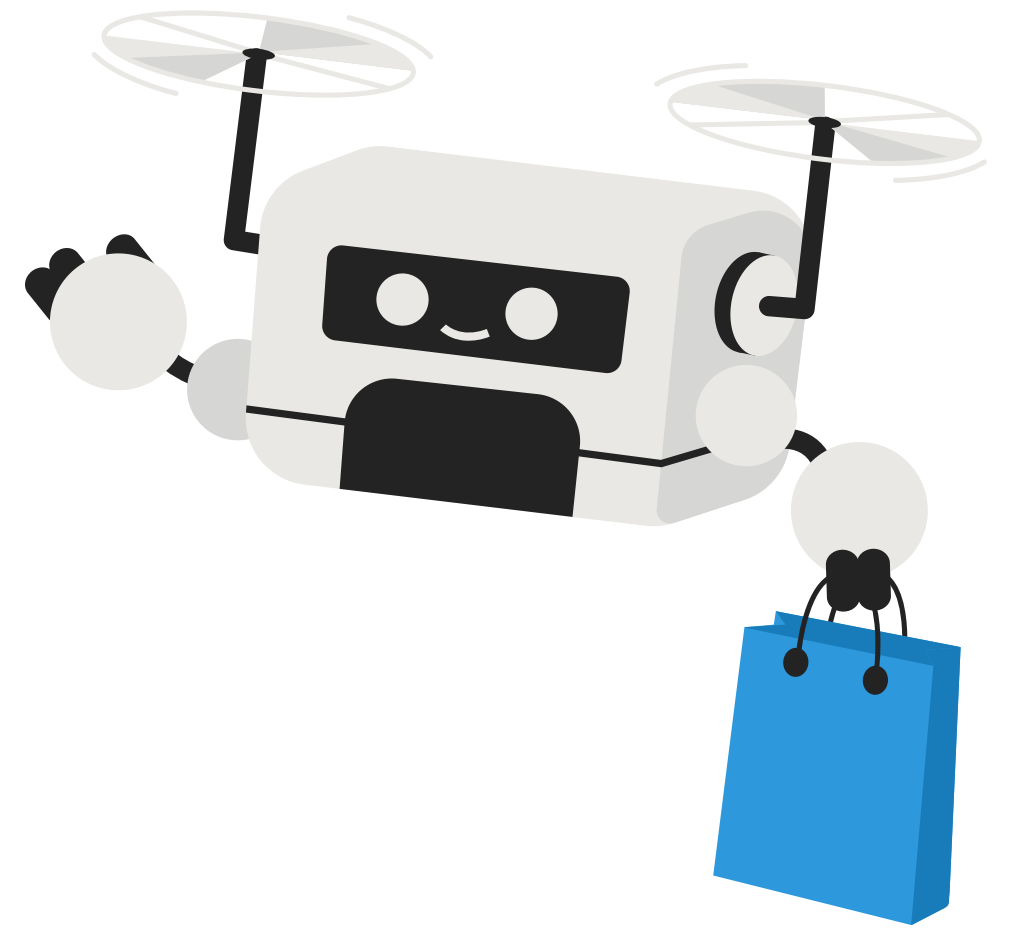
Command: SET_POSITION_TARGET_LOCAL_NED

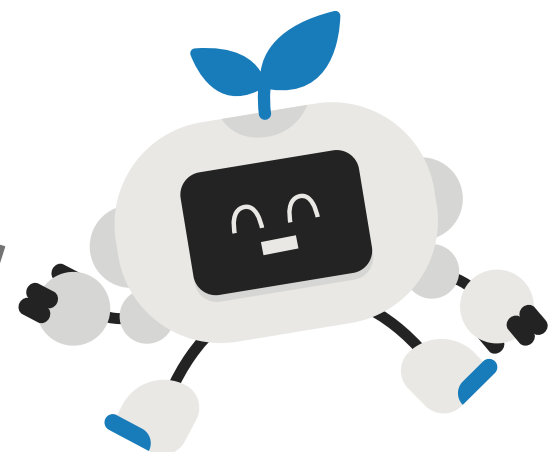
Target: X, Y, Z Position (Body Frame Offset)



Real-Time visualization

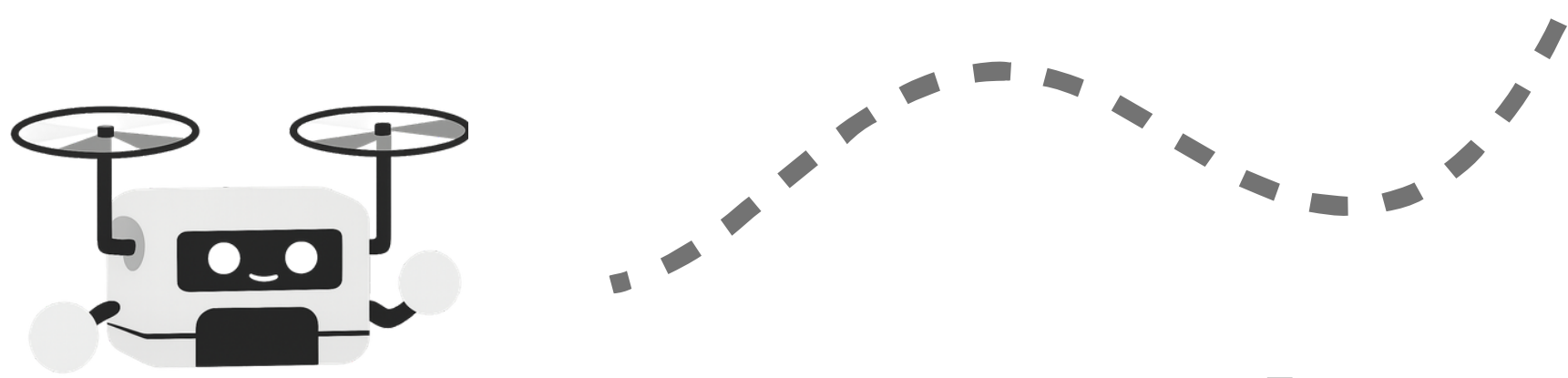
- ✓ 3D visualization using RViz2
- ✓ Robot State Publisher +
tf_broadcates.py





RESULTS AND CONCLUSION





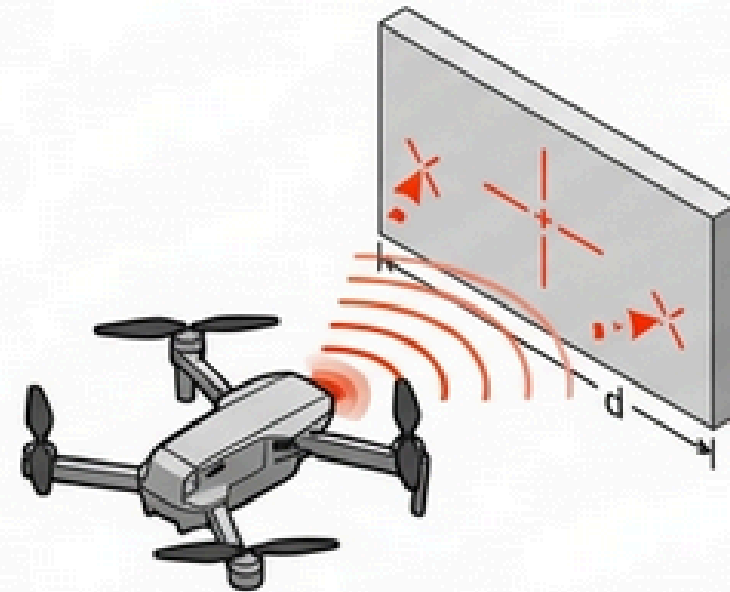
Future Work

Physical Implementation



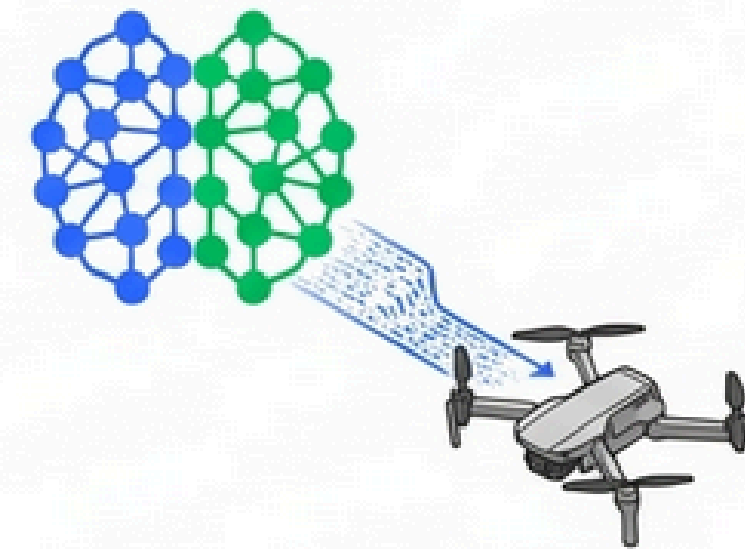
Physical Implementation

Forward Collision Avoidance



Forward Collision Avoidance

AI Integration



CNN Object Tracking



**Thank You
Very Much!**

